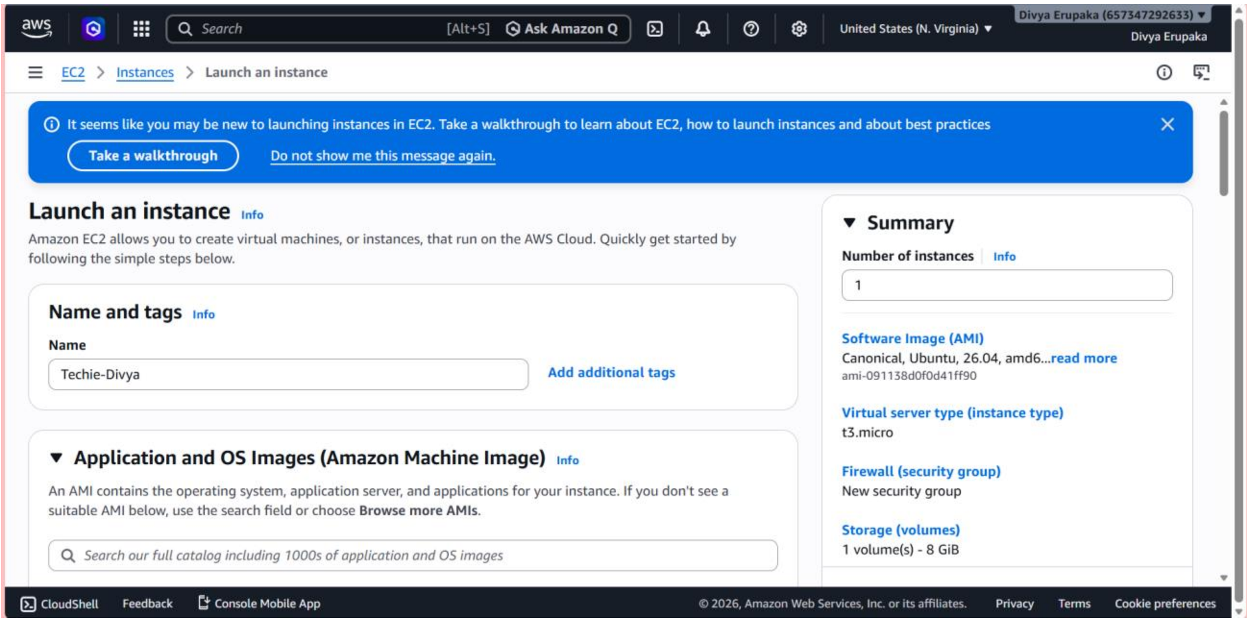
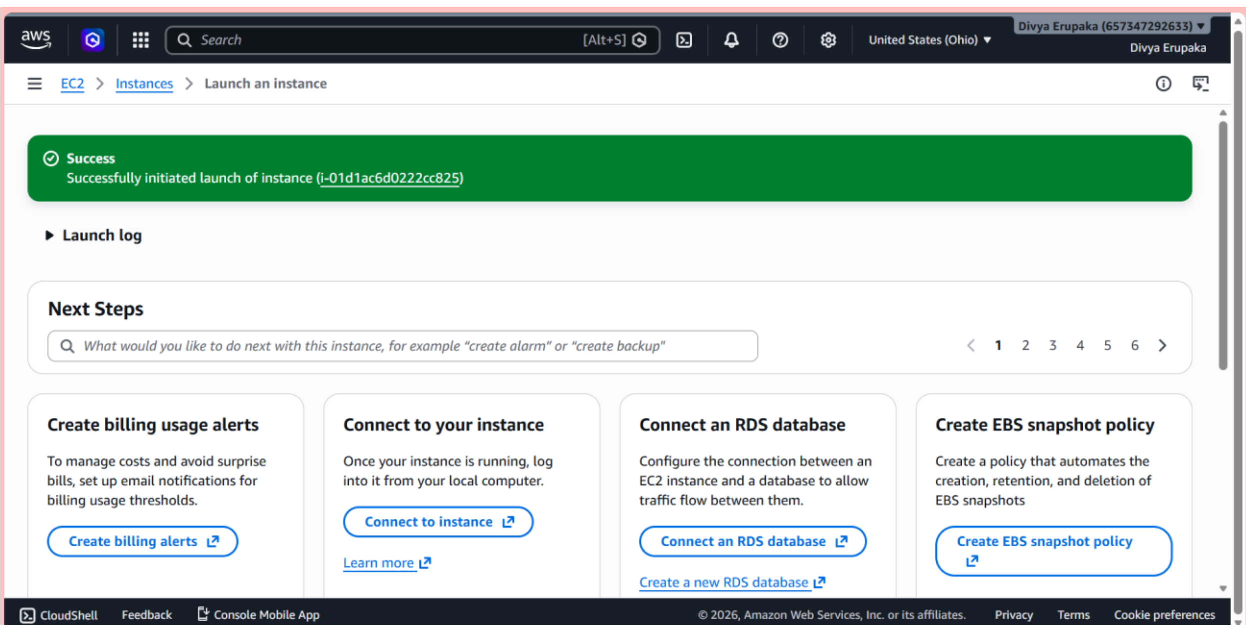
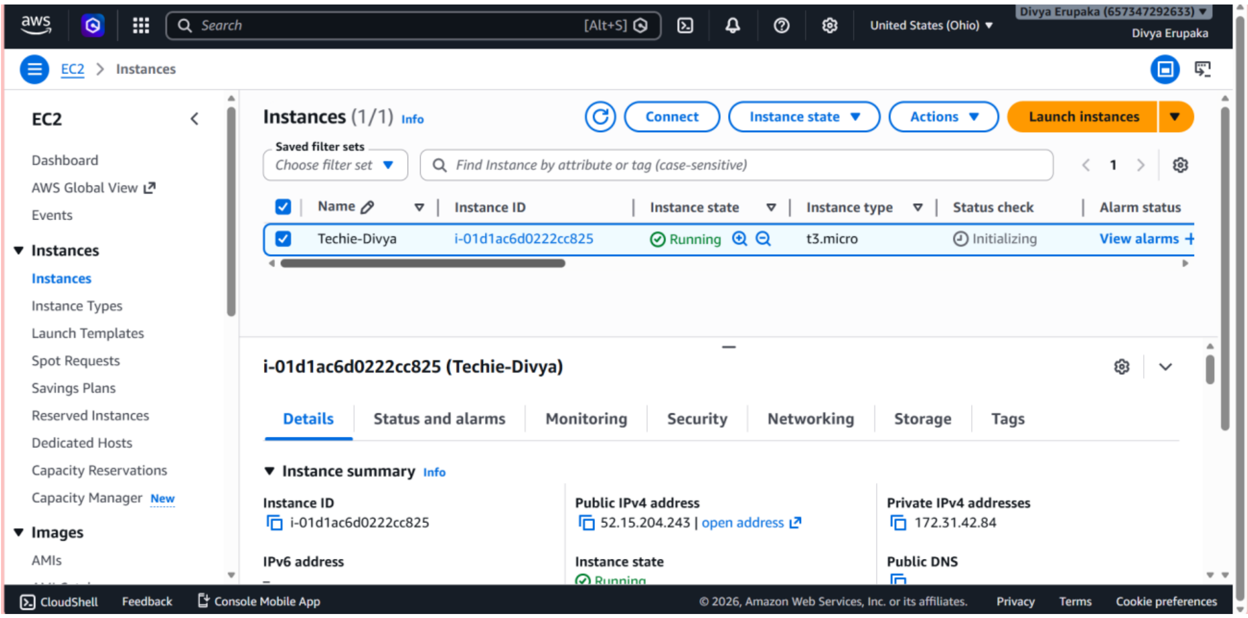
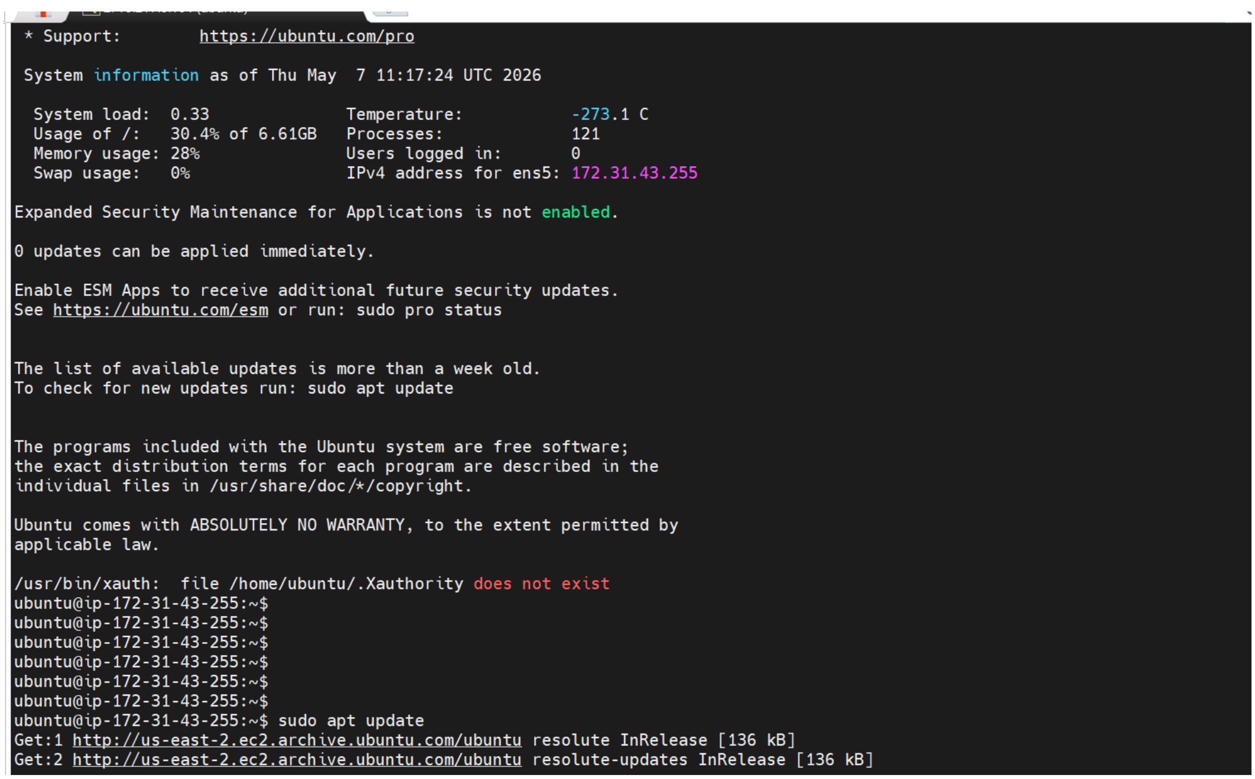
EC2 Instance Launch







Connected to Instance



```
ubuntu@ip-172-31-43-255:~$ sudo apt install openjdk-17-jdk -y
Installing:
  openjdk-17-jdk

Installing dependencies:
  adwaita-icon-theme          libatk-bridge2.0-0t64    libglx0                libsm6              libxmu6
  alsa-topology-conf          libatk-wrapper-java      libglycin-2-0          libthai-data        libxpm4
  alsa-ucm-conf               libatk-wrapper-java-jni  libgraphite2-3         libthai0            libxrandr2
  at-spi2-common              libatk1.0-0t64           libgtk2.0-0t64         libvulkan1          libxrender1
  at-spi2-core                libatspi2.0-0t64         libgtk2.0-bin          libwayland-client0  libxres1
  bubblewrap                  libavahi-client3         libgtk2.0-common       libx11-dev          libxshmfence1
  ca-certificates-java        libavahi-common-data     libharfbuzz0b          libx11-xcb1         libxt-dev
  dconf-gsettings-backend     libavahi-common3         libheif-plugin-aomdec  libxau-dev          libxt6t64
  dconf-service               libc-dev-bin             libheif-plugin-aomenc  libxaw7             libxtst6
  fontconfig                  libc6-dev                libheif1               libxcb-dri3-0       libxv1
  fontconfig-config           libcairo-gobject2        libhwy1t64             libxcb-glx0         libxxf86dga1
  fonts-dejavu-core           libcairo2                libibus-1.0-5          libxcb-present0     libxxf86vm1
  fonts-dejavu-extra          libcups2t64              libice-dev             libxcb-randr0       linux-libc-dev
  fonts-dejavu-mono           libdatrie1               libice6                libxcb-render0      luit
  glycin-loaders              libdconf1                libjpeg-turbo8         libxcb-shape0       manpages-dev
  glycin-thumbnailers         libdisplay-info3         libjpeg8               libxcb-shm0         mesa-libgallium
  gnome-accessibility-themes  libdrm-intel1            libjxl0.11             libxcb-sync1        mesa-vulkan-drivers
  gnome-themes-extra          libfontconfig1           liblcms2-2             libxcb-xfixes0      openjdk-17-jdk-headless
  gnome-themes-extra-data     libgail-common           libpango-1.0-0         libxcb1-dev         openjdk-17-jre
  gsettings-desktop-schemas   libgail18t64             libpangocairo-1.0-0    libxcomposite1      openjdk-17-jre-headless
  gtk-update-icon-cache       libgbm1                  libpangoft2-1.0-0      libxcursor1         rpcsvc-proto
  gtk2-engines-pixbuf         libgdk-pixbuf-2.0-0      libpciaccess0          libxdamage1         user-session-migration
  hicolor-icon-theme          libgdk-pixbuf2.0-common  libpcsclite1           libxdmcp-dev        uuid-dev
  ibus-gtk                    libgif7                  libpixman-1-0          libxfixes3          x11-common
  java-common                 libgl1                   librsvg2-2             libxft2             x11-utils
  libaom3                     libgl1-mesa-dri          librsvg2-common        libxi6              x11proto-dev
  libasound2-data             libglvnd0                libsharpyuv0           libxinerama1        xorg-sgml-doctools
  libasound2t64               libglx-mesa0             libsm-dev              libxkbfile1         xtrans-dev

Suggested packages:
```

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-43-255:~$ java -version
openjdk version "17.0.18" 2026-01-20
OpenJDK Runtime Environment (build 17.0.18+8-Ubuntu-1)
OpenJDK 64-Bit Server VM (build 17.0.18+8-Ubuntu-1, mixed mode, sharing)
ubuntu@ip-172-31-43-255:~$
```

Attempted Jenkins Repository Installation

Initially we tried installing Jenkins using Jenkins repository.

Commands included:

- Adding Jenkins repo
- Importing GPG keys
- Running apt update

Problem faced:

- Ubuntu version was resolute (new Ubuntu release)
- Jenkins repository did not officially support that version yet
- Repository signature verification failed

Error seen:

NO_PUBKEY 7198F4B714ABFC68

Alternative Method — Install Jenkins Using .deb Package

Instead of repository installation, we downloaded Jenkins package manually.

Command used:

wget https://pkg.jenkins.io/debian-stable/binary/jenkins_2.504.1_all.deb

Purpose:

- Downloads Jenkins Debian package directly from Jenkins official website.

Downloaded file:

- jenkins_2.504.1_all.deb

Install Jenkins Package

Command used:

sudo dpkg -i jenkins_2.504.1_all.deb

Purpose:

- dpkg installs .deb packages in Ubuntu/Debian systems.
- Jenkins application files and service files were installed.

What happened internally:

- Jenkins user got created
- Jenkins service got registered with systemd
- Jenkins files got placed in:
    - /var/lib/jenkins
    - /etc/systemd/system
    - /usr/share/jenkins

Fix Broken Dependencies (if required)

Command used:

sudo apt --fix-broken install -y

Purpose:

- Automatically installs missing dependencies required by Jenkins.

Start Jenkins Service

Command used:

sudo systemctl start jenkins

Purpose:

- Starts Jenkins service immediately.

Internally:

- Jenkins Java process starts running in background.

Step 10: Enable Jenkins Service

Command used:

sudo systemctl enable jenkins

Purpose:

- Enables Jenkins to start automatically whenever server reboots.

Step 11: Verify Jenkins Status

Command used:

sudo systemctl status jenkins

Purpose:

- Checks whether Jenkins service is running properly.

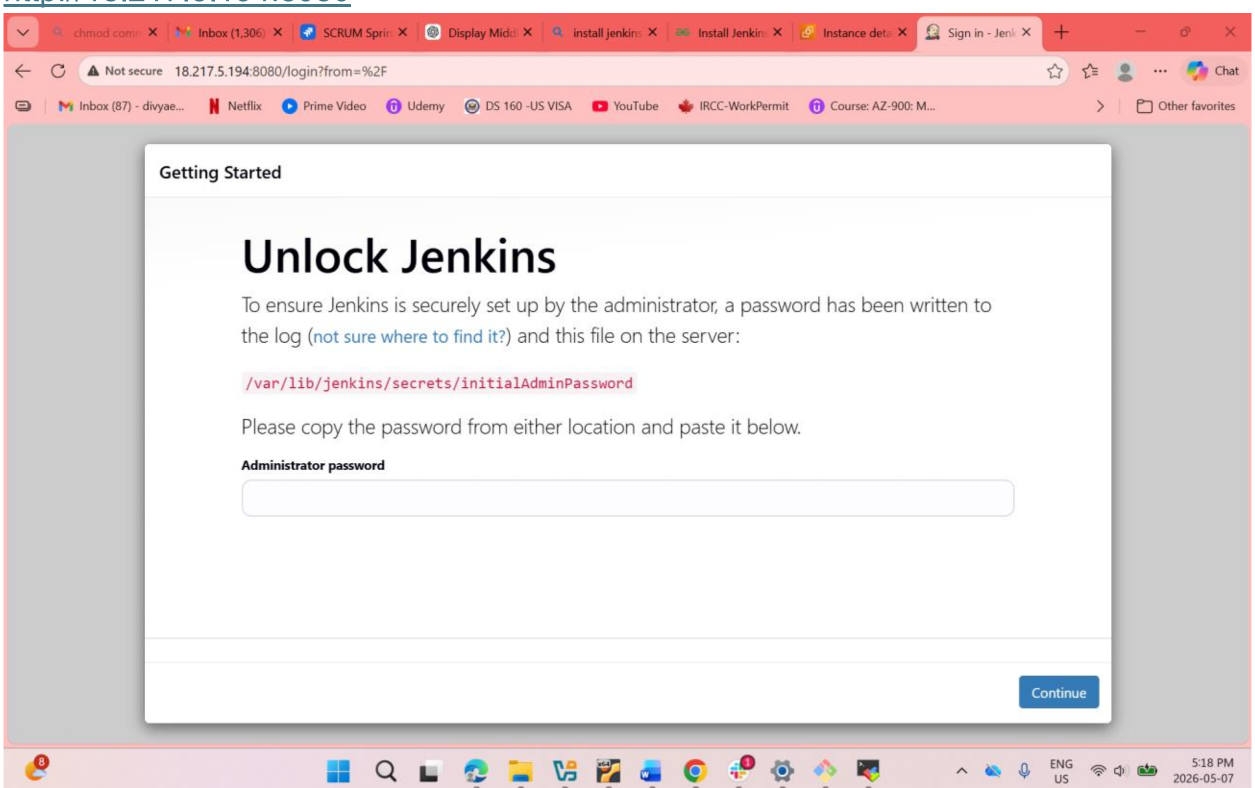
Expected output:

active (running) ---- Jenkins server is successfully running.

```
(Reading database ... 98958 files and directories currently installed.)
Preparing to unpack .../net-tools_2.10-2ubuntu1_amd64v3.deb ...
Unpacking net-tools (2.10-2ubuntu1) ...
Setting up net-tools (2.10-2ubuntu1) ...
Setting up jenkins (2.504.1) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/jenkins.service' → '/usr/lib/systemd/system/jenkins.service'.
Processing triggers for man-db (2.13.1-1build1) ...
needrestart is being skipped since dpkg has failed
root@ip-172-31-43-255:~#
root@ip-172-31-43-255:~#
root@ip-172-31-43-255:~#
root@ip-172-31-43-255:~# sudo systemctl start jenkins
sudo systemctl enable jenkins
Synchronizing state of jenkins.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable jenkins
root@ip-172-31-43-255:~# sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-05-07 11:39:14 UTC; 32s ago
 Invocation: 077c8461e3374c2da6f1282d8c00e64b
   Main PID: 6345 (java)
      Tasks: 43 (limit: 627)
     Memory: 310.4M (peak: 343.7M)
        CPU: 23.218s
     CGroup: /system.slice/jenkins.service
             └─6345 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war >

May 07 11:39:09 ip-172-31-43-255 jenkins[6345]: This may also be found at: /var/lib/jenkins/secrets/initialAdminPassword
May 07 11:39:09 ip-172-31-43-255 jenkins[6345]: *************************************************************
May 07 11:39:09 ip-172-31-43-255 jenkins[6345]: *************************************************************
May 07 11:39:09 ip-172-31-43-255 jenkins[6345]: *************************************************************
May 07 11:39:14 ip-172-31-43-255 jenkins[6345]: 2026-05-07 11:39:14.781+0000 [id=33]        INFO        jenkins.InitReactorRu>
May 07 11:39:14 ip-172-31-43-255 jenkins[6345]: 2026-05-07 11:39:14.811+0000 [id=23]        INFO        hudson.lifecycle.Life>
May 07 11:39:14 ip-172-31-43-255 systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
May 07 11:39:15 ip-172-31-43-255 jenkins[6345]: 2026-05-07 11:39:15.089+0000 [id=49]        INFO        h.m.DownloadService$D>
May 07 11:39:15 ip-172-31-43-255 jenkins[6345]: 2026-05-07 11:39:15.090+0000 [id=49]        INFO        hudson.util.Retrier#s>
May 07 11:39:19 ip-172-31-43-255 jenkins[6345]: 2026-05-07 11:39:19.904+0000 [id=68]        WARNING        h.n.DiskSpaceMonit>
lines 1-21/21 (END)
```

URL used:

http://18.217.5.194:8080




```
root@ip-172-31-43-255:~# sudo cat /var/lib/jenkins/secrets/initialAdminPassword
13623dca2fd245a1bab805d01a196a77
```

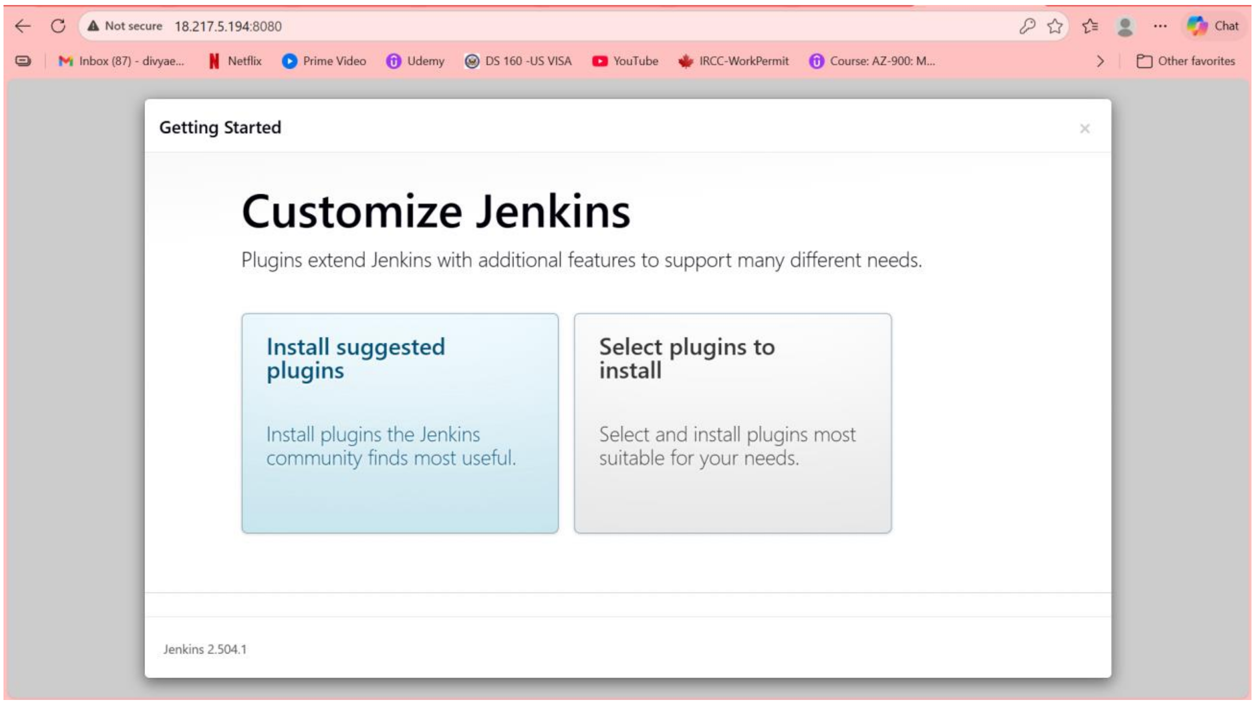
Retrieve Initial Admin Password

Command used:

sudo cat /var/lib/jenkins/secrets/initialAdminPassword

Purpose:

- Jenkins generates a temporary admin password during first startup.
- Needed for first-time login.



Tomcat Installation in Ubuntu VM

I have checked the tomcat in repository using command
sudo apt-cache search tomcat

confirmed that tomcat10 is available

```
ubuntu@ip-172-31-43-255:~$ sudo apt-cache search tomcat
centreon-plugins - Collection of Nagios plugins to monitor OS, services and network devices
css-validator-tomcat10 - W3C CSS Validator - tomcat10 configuration files
css-validator-tomcat11 - W3C CSS Validator - tomcat11 configuration files
libapache-mod-jk-doc - Documentation of libapache2-mod-jk package
libapache2-mod-jk - Apache 2 connector for the Tomcat Java servlet engine
libjnlp-servlet-java - simple and convenient packaging format for JNLP applications
liblogback-java - flexible logging library for Java
libnetty-tcnative-java - Tomcat native fork for Netty
libnetty-tcnative-jni - Tomcat native fork for Netty (JNI library)
libspring-instrument-java - modular Java/J2EE application framework - Instrumentation
libtcnative-1 - Tomcat native library using the Apache Portable Runtime
libtomcat10-embed-java - Apache Tomcat 10 - Servlet and JSP engine -- embed libraries
libtomcat10-java - Apache Tomcat 10 - Servlet and JSP engine -- core libraries
libtomcat11-embed-java - Apache Tomcat 11 - Servlet and JSP engine -- embed libraries
libtomcat11-java - Apache Tomcat 11 - Servlet and JSP engine -- core libraries
libtomcat9-java - Apache Tomcat 9 - Servlet and JSP engine -- core libraries
monitoring-plugins-contrib - Plugins for nagios compatible monitoring systems
python3-ajpy - Python module to craft AJP requests
resource-agents-extra - Cluster Resource Agents
tomcat-jakartaee-migration - Apache Tomcat migration tool for Jakarta EE
tomcat10 - Apache Tomcat 10 - Servlet and JSP engine
tomcat10-admin - Apache Tomcat 10 - Servlet and JSP engine -- admin web applications
tomcat10-common - Apache Tomcat 10 - Servlet and JSP engine -- common files
tomcat10-docs - Apache Tomcat 10 - Servlet and JSP engine -- documentation
tomcat10-examples - Apache Tomcat 10 - Servlet and JSP engine -- example web applications
tomcat10-user - Apache Tomcat 10 - Servlet and JSP engine -- tools to create user instances
tomcat11 - Apache Tomcat 11 - Servlet and JSP engine
tomcat11-admin - Apache Tomcat 11 - Servlet and JSP engine -- admin web applications
tomcat11-common - Apache Tomcat 11 - Servlet and JSP engine -- common files
tomcat11-docs - Apache Tomcat 11 - Servlet and JSP engine -- documentation
tomcat11-examples - Apache Tomcat 11 - Servlet and JSP engine -- example web applications
tomcat11-user - Apache Tomcat 11 - Servlet and JSP engine -- tools to create user instances
yasat - simple stupid audit tool
ubuntu@ip-172-31-43-255:~$ sudo apt install tomcat9 tomcat9-admin
```

Install Tomcat 10

Using command ---- sudo apt install tomcat10 tomcat10-admin -y

Start Tomcat service:

sudo systemctl start tomcat10

Enable Tomcat service:

sudo systemctl enable tomcat10

Check Tomcat status:

sudo systemctl status tomcat10

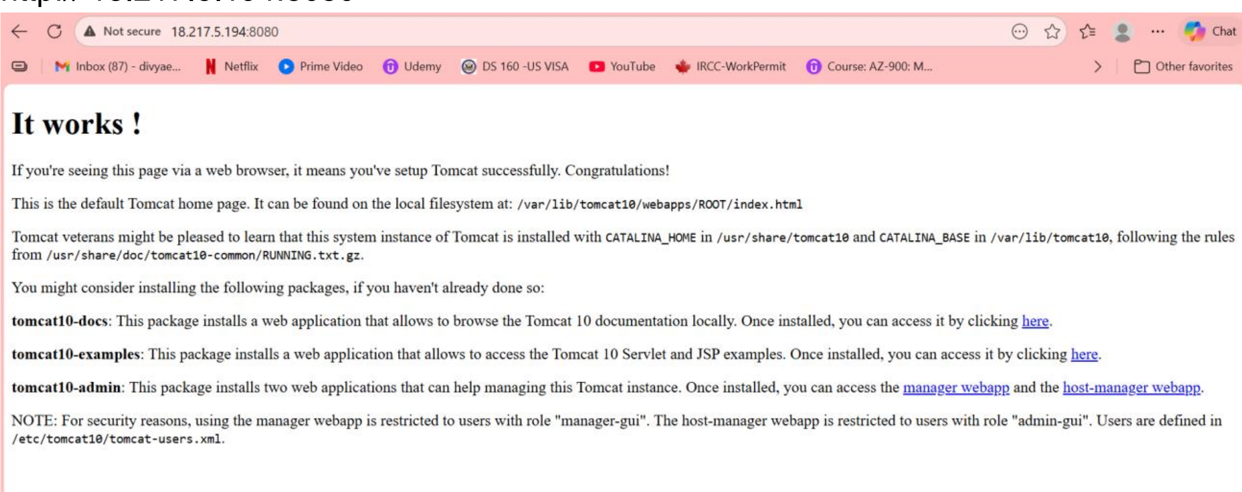
If status shows: active (running)

then Tomcat is running successfully.

```
No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-43-255:~$
ubuntu@ip-172-31-43-255:~$
ubuntu@ip-172-31-43-255:~$
ubuntu@ip-172-31-43-255:~$
ubuntu@ip-172-31-43-255:~$ sudo systemctl start tomcat10
ubuntu@ip-172-31-43-255:~$ sudo systemctl enable tomcat10
ubuntu@ip-172-31-43-255:~$ sudo systemctl status tomcat10
● tomcat10.service - Apache Tomcat 10 Web Application Server
     Loaded: loaded (/usr/lib/systemd/system/tomcat10.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-05-07 12:11:29 UTC; 30s ago
 Invocation: 10cd95c882e04e2a9558d0aa6155c861
       Docs: https://tomcat.apache.org/tomcat-10.0-doc/index.html
   Main PID: 8029 (java)
      Tasks: 18 (limit: 627)
     Memory: 94.2M (peak: 118.2M)
        CPU: 8.932s
     CGroup: /system.slice/tomcat10.service
             └─8029 /usr/lib/jvm/java-17-openjdk-amd64/bin/java -Djava.util.logging.config.file=/var/lib/tomcat10/conf/loggin>

May 07 12:11:34 ip-172-31-43-255 tomcat10[8029]: At least one JAR was scanned for TLDs yet contained no TLDs. Enable debug lo>
May 07 12:11:34 ip-172-31-43-255 tomcat10[8029]: Deployment of deployment descriptor [/etc/tomcat10/Catalina/localhost/host-m>
May 07 12:11:34 ip-172-31-43-255 tomcat10[8029]: Deploying deployment descriptor [/etc/tomcat10/Catalina/localhost/manager.xm>
May 07 12:11:34 ip-172-31-43-255 tomcat10[8029]: The path attribute with value [/manager] in deployment descriptor [/etc/tomc>
May 07 12:11:35 ip-172-31-43-255 tomcat10[8029]: At least one JAR was scanned for TLDs yet contained no TLDs. Enable debug lo>
May 07 12:11:35 ip-172-31-43-255 tomcat10[8029]: Deployment of deployment descriptor [/etc/tomcat10/Catalina/localhost/manage>
May 07 12:11:35 ip-172-31-43-255 tomcat10[8029]: Deploying web application directory [/var/lib/tomcat10/webapps/ROOT]
May 07 12:11:36 ip-172-31-43-255 tomcat10[8029]: At least one JAR was scanned for TLDs yet contained no TLDs. Enable debug lo>
May 07 12:11:36 ip-172-31-43-255 tomcat10[8029]: Deployment of web application directory [/var/lib/tomcat10/webapps/ROOT] has>
May 07 12:11:36 ip-172-31-43-255 tomcat10[8029]: Server startup in [4808] milliseconds
lines 1-22/22 (END)
```

access Tomcat in browser:

http:// 18.217.5.194:8080



Jenkins is already using port 8080

> Tomcat also uses port 8080 by default . So Tomcat may failed to start due to port conflict

To stop Jenkins: sudo systemctl stop Jenkins
Then restart Tomcat: sudo systemctl restart tomcat10

Setup Docker Desktop on your local windows.